# Unseen Steps: prepared like a describer, delivered live

A studio's rehearsal becomes show notes; on the night, Gemini Live narrates what the dancer actually does, in the gaps in the music.

## Before the show

from the studio's rehearsal video

### Rehearsal video + song

Uploaded by the studio. librosa maps the music: beats, phrases, energy, and the quiet gaps where a line fits.

### Gemini 4 Argon

✓ every line checked against the video at its moment (10 fps)

1. Performance map of the whole piece at 5 fps: style with evidence, sections, signature, texture changes, key moments
2. Re-watches each section at 8 fps
3. Writes a short stretch at a time with a dancer's-eye brief and a dancer's feedback
4. Second-describer review

### Show notes + rehearsal script

Style, signature, sections, key moments, the song's gaps, and timed lines in the style's own vocabulary.

### MongoDB Atlas

**shows**: stored per show; the studio edits them (scripts/edit_show.py). **Atlas Search** finds a song by name. **Hybrid Search** (Vector Search + Atlas Search) matches movement to a dancer-vetted glossary.

## During the show

a camera on the stage, the same song

### Camera

Follows the dancer. Frames go to Gemini at 2 per second.

### YOLO11 pose

Measures jumps, turns and their direction, facing, footwork, floor work, turnout. Unsure joints are left out; unmeasurable checks are skipped, never guessed.

### Gemini Live (gemini-3.8-live)

Holds the show notes and knows where it is in the song. In each gap it describes what the dancer does now, including anything different from rehearsal. First sound about 0.8 s after the prompt.

### Pose checker

✓ live verification

Compares every claim with the measurements. A contradicted line is flagged, Gemini is told, and the next line puts it right.

### Listener's earphones

Each blind or partially sighted audience member, on their own phone; never over the room's music.

**Recorded narration** (the 18 style demos, the ballet solo): gemini-3.8-flash-tts speaks each line; gemini-3.8-flash transcribes it and it must say exactly its caption, or it is re-recorded or dropped. ✓ caption check

**Privacy**: live frames are not stored unless the studio records; in recordings everyone except the dancers being described is blurred. **Honesty**: if tracking is unsure, it measures nothing and says less.

Open source (MIT): github.com/anushaisawesome/unseen-steps · Try it: unseen-steps-studio.vercel.app

✓ a verification check